

Street trees improve the city environment, so they are important to urban planning. One of the most important benefits is that they create shade. Therefore, in this analysis, we will look at what types of trees work well to maximize shade so that the city can efficiently maximize the amount of shade produced by the trees in Raleigh.

First, we will look at the sample size of our data, as a small sample size may negatively affect the analysis. Arizona has 827 observations, California has 4062 observations, Colorado has 867 observations, Florida has 895 observations, Hawaii has 918 observations, Idaho has 923 observations, Indiana has 877 observations, Minnesota has 760 observations, North Carolina has 828 observations, New Mexico has 833 observations, New York has 831 observations, South Carolina has 872 observations, and Washington has 994 observations. Figure 1 shows the sample size by state in a bar graph.

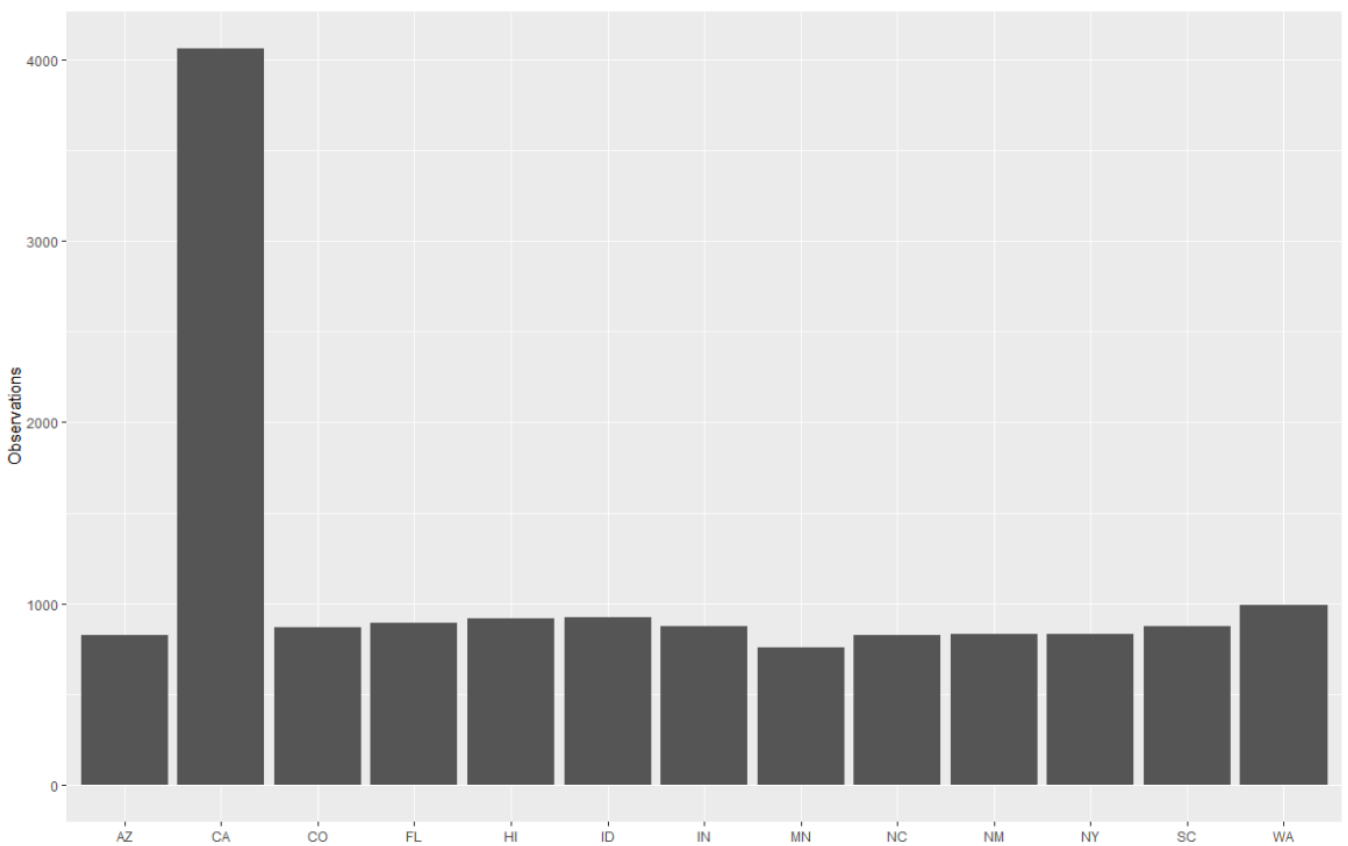


Figure 1
Sample size by state

Although a bigger dataset makes the analysis more accurate, we shouldn't use all the data in this dataset. Since trees are living things, they grow differently in different areas, and different trees grow in different areas. Therefore, for this analysis, we will only use data from North Carolina and South Carolina. The data for North Carolina was collected in Charlotte, and the data for South Carolina was collected in Charleston. Since we want to maximize the shade area, we will look at the average crown size for each genus. The Quercus genus had the

biggest crown size at 13.623163 meters. Figure 2 is a bar graph showing the average crown size by genus.

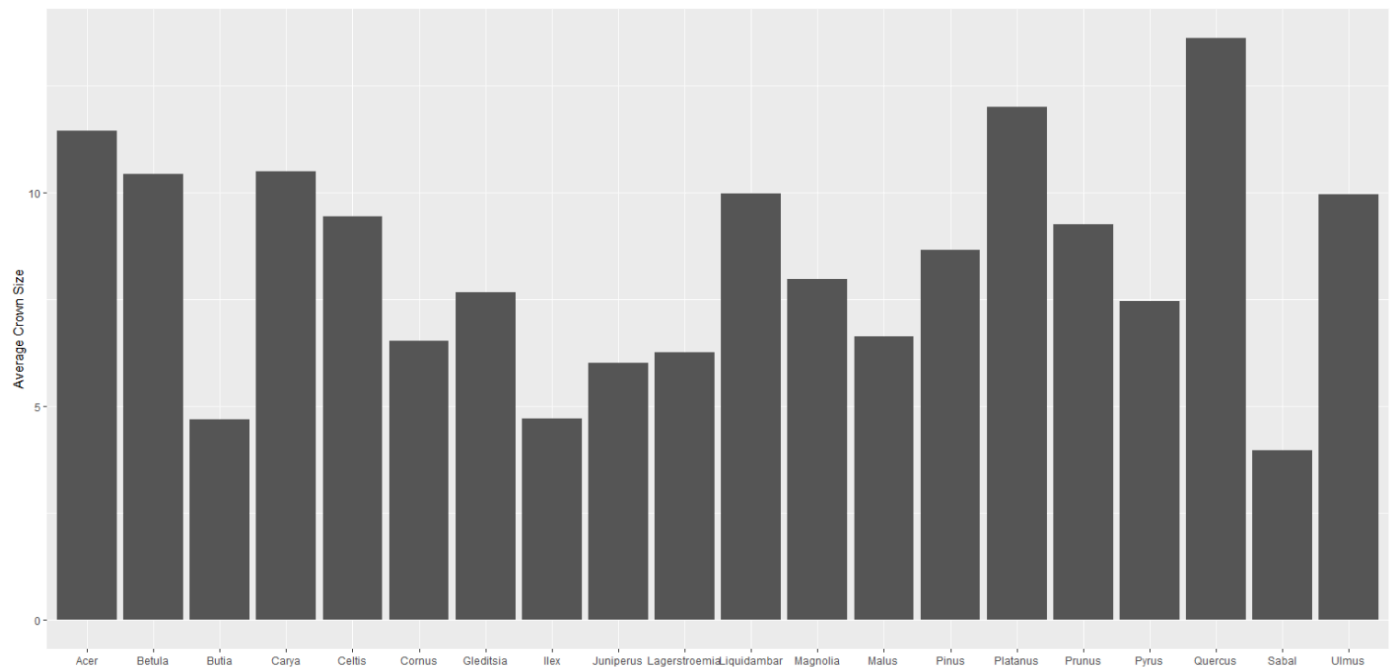


Figure 2
Average crown size by genus

However, this does not mean that the Quercus genus is the best genus to plant. The bigger crown sizes may simply be due to higher average tree age. Therefore, we will now look at the average ages of each genus, and its relation with the average crown size. Figure 3 shows the scatter plot of the average age on the horizontal axis and the average crown size on the vertical axis.

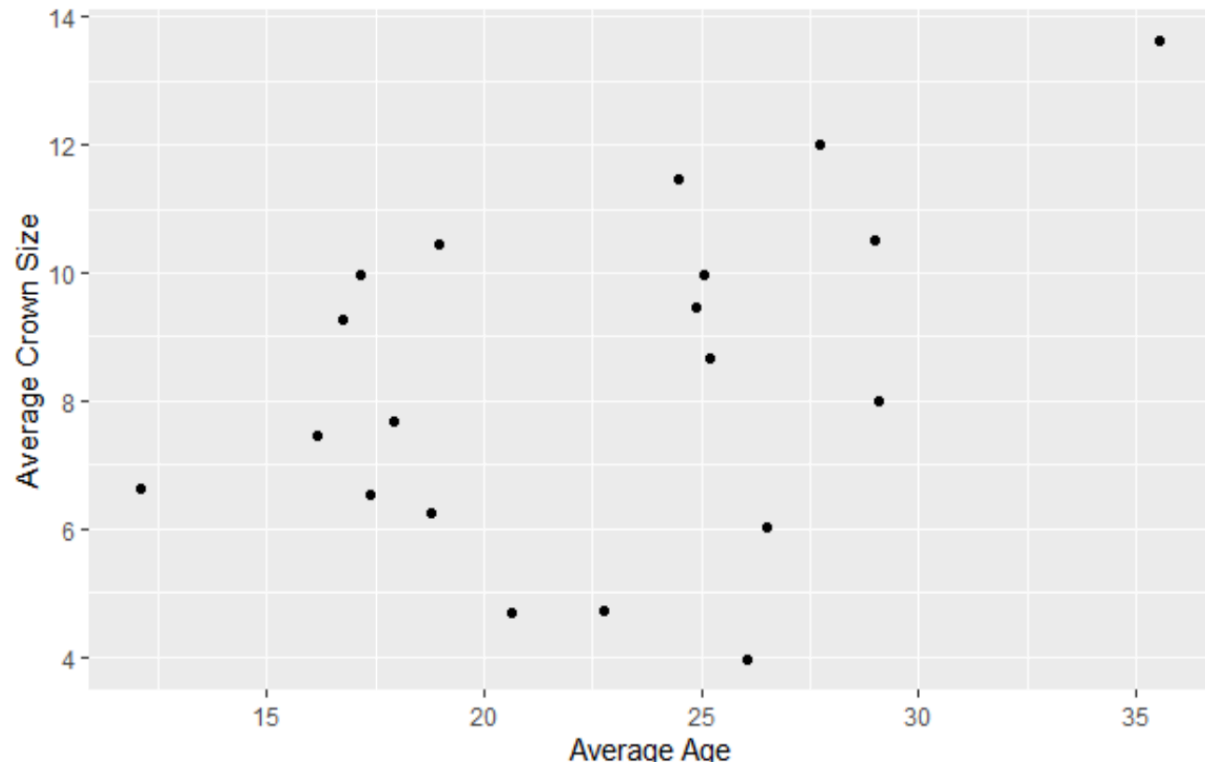


Figure 3
Average crown size vs average tree age

Although there is a positive correlation between the average tree age and the average crown size, the correlation is very low because the R^2 is 0.1208. It is also worth calculating the efficiency of the genus. In this analysis, I will be calculating the effectiveness by dividing the average crown size by the average tree age; this number explains how much the tree's crown size grew in a year on average. Figure 4 shows these numbers in a bar graph.

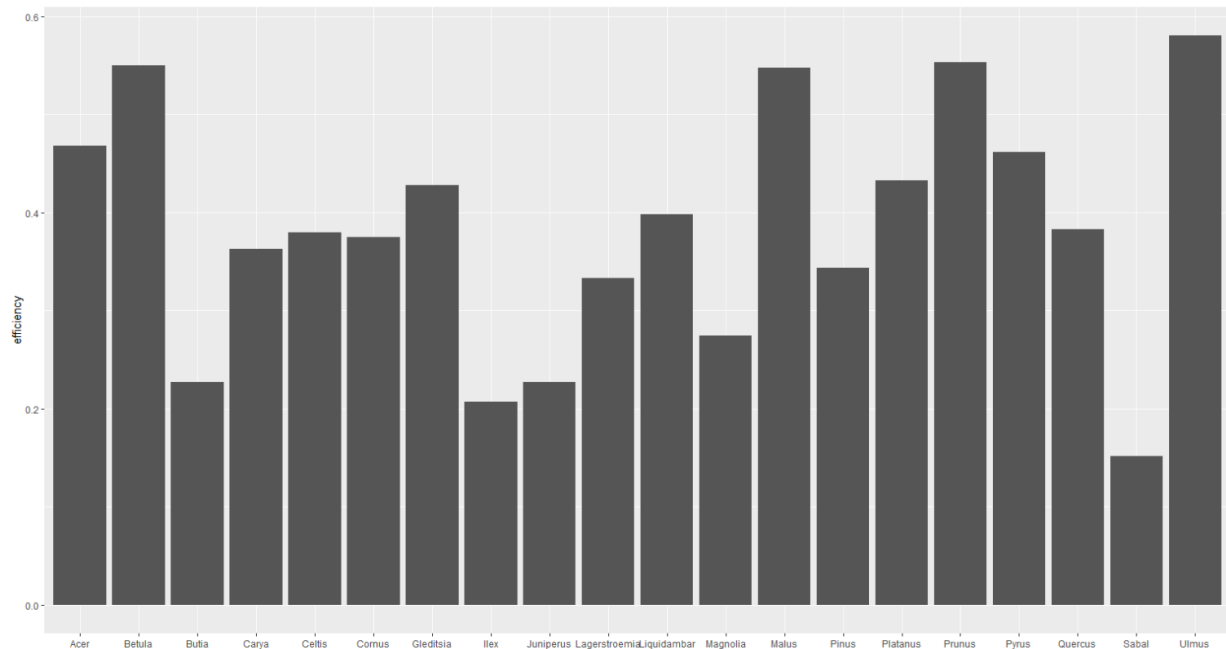


Figure 4
Efficiency of each genus in North or South Carolina

The Ulmus genus is the most efficient, with an average of a 0.5806922 meter growth of the crown size per year. Therefore, planting trees of the Ulmus genus is recommended. However, this may simply be due to the tree genus's relatively low average age. At a certain age, trees will stop growing rapidly, so older trees will stay a certain size for a long time. Therefore, we would be able to conduct a more accurate analysis if we had the average age of when each tree genus reaches adulthood.

AI reflection

For this specific homework, I haven't used any AI, but I did use some outside sources. I had trouble with correctly separating the state and the city because there were some cities that contained two words, such as Fort Collins, CO. From R's help website, I was able to figure out how to match anything except a comma and properly separate the city and state.